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from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, classification_report, roc_auc_score,
    roc_curve
)
from sklearn.datasets import load_breast_cancer
import pandas as pd
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore", category=FutureWarning)
data = load_breast_cancer()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target, name="target")
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42, stratify=y)

scaler = StandardScaler()
X_train_sc = scaler.fit_transform(X_train)
X_test_sc = scaler.transform(X_test)
svm = SVC(kernel='rbf', probability=True, random_state=42)
param_grid = {
    "C": [0.1, 1, 10, 100],
    "gamma": ["scale", 0.01, 0.001, 0.0001],
}
grid = GridSearchCV(estimator=svm, param_grid=param_grid, scoring='f1', cv=5, n_jobs=1, verbose=0)
grid.fit(X_train_sc, y_train)
print("Best Parameters from Grid Search:", grid.best_params_)
best_svm = grid.best_estimator_
best_svm.fit(X_train_sc, y_train)
y_pred = best_svm.predict(X_test_sc)
y_prob = best_svm.predict_proba(X_test_sc)[:, 1]
acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred, zero_division=0)
rec = recall_score(y_test, y_pred)

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```
print(f"Accuracy : {acc:.4f}")
print(f"Precision: {prec:.4f}")
print(f"Recall : {rec:.4f}")
print(f"F1-Score : {f1:.4f}")
print(f"ROC-AUC : {auc:.4f}")
print("\nConfusion Matrix:\n", cm)
print("\nClassification Report:\n", classification_report(y_test, y_pred, zero_division=0))
fpr, tpr, thresholds = roc_curve(y_test, y_prob)
plt.figure()
plt.plot(fpr, tpr, label=f"SVM (AUC = {auc:.3f})")
plt.plot([0, 1], [0, 1], linestyle="--", color='gray')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - SVM (RBF)")
plt.legend()
plt.grid(True)
plt.show()
```

Best Parameters from Grid Search: {'C': 10, 'gamma': 0.01}

=== SVM (RBF) - Test Metrics ===

Accuracy : 0.9825
Precision: 0.9861
Recall : 0.9861
F1-Score : 0.9861
ROC-AUC : 0.9977

Confusion Matrix:

```
[[41  1]
 [ 1 71]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	42
1	0.99	0.99	0.99	72
accuracy			0.98	114
macro avg	0.98	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

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ROC Curve — SVM (RBF)

